

KUBERNETES IN PRODUCTION · ISSUE 05 OF 08

Namespaces and RBAC for multi-team clusters.

A namespace is a boundary only if RBAC, quotas and policy agree.

05

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Akhilesh Ranjan Singh

Senior DevSecOps Engineer · github.com/iarsingh

Swipe →

Namespace design

01 One per team and environment

payments-dev, payments-prod — clear ownership.

02 ResourceQuota

One team cannot starve the cluster.

03 LimitRange

Sensible defaults when a pod sets nothing.

04 Owner labels

owner and cost-center on every namespace.



RBAC mistakes

DO

- ✓ Role over ClusterRole
- ✓ Bind groups from your IdP
- ✓ One deploy identity per namespace
- ✓ automountServiceAccountToken: false

AVOID

- ✗ cluster-admin for CI
- ✗ Wildcard verbs and resources
- ✗ Broad get/list on secrets
- ✗ Default SA tokens everywhere

Who gets what

WHO	ACCESS
Developers	View + logs in own namespace
CI deployer	Apply in own namespace only
Platform team	Admin via audited break-glass
Bots and agents	Read-only unless reviewed



Audit in two commands

BASH

```
kubectl auth can-i --list -n payments \
  --as=system:serviceaccount:payments:ci

kubectl get clusterrolebindings -o wide \
  | grep cluster-admin
```

Run this quarterly. Bindings only ever grow.



Shared cluster baseline

- ☒ Quota and LimitRange per namespace

- ☒ Default-deny NetworkPolicy

- ☒ No cluster-admin bindings for CI

- ☒ Secrets readable only by workloads

- ☒ Pod Security Admission: restricted

- ☒ Audit logs shipped off-cluster



KEY TAKEAWAY

Tenancy is quotas, RBAC and network policy together.

- Namespace per team and env
- Least-privilege, group-based RBAC
- Audit bindings regularly

YOUR TURN

Who has cluster-admin in your cluster right now — and do you know why?

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